

Daniel Medeiros

Researcher | Applied AI & HPC Specialist

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- Location: Stockholm, Sweden
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Short Profile

Systems-oriented researcher specializing in data-intensive systems, cloud computing, and applied AI. I bridge the gap between academic research and industrial application, with a track record in High-Performance Computing (HPC) and large-scale geospatial engineering. Currently driving AI adoption for SMEs and the public sector via EuroHPC initiatives at RISE.

Work Experience

- **2025 - Present: Researcher @ RISE Research Institutes of Sweden**
 - **Technical Consultancy:** Supporting SMEs and the public sector under EU-funded projects (ENCCS and MIMER AI Factory) with AI implementation, HPC onboarding, and scalable proof-of-concepts.
 - **Business Development:** Managing lead pipelines via Pipedrive (CRM) and driving collaborations through startup incubators and Swedish AI events.
 - **Training:** Organizing, supporting and/or delivering technical workshops on GPU Programming, Cloud Computing, Containerization, and Distributed Systems.
- **2014 - 2021: Geophysicist @ Geological Survey of Brazil**
 - **Automation:** Developed Python-based software and plugins for ArcGIS and Oasis montaj using libraries such as GDAL and GeoPandas to automate manual data workflows.
 - **Data Analysis:** Managed acquisition and processing of aerogeophysical data (gravimetric, magnetic, spectrometric) for mineral exploration.
 - **Seismic & QC:** Performed 2D seismic processing with SeisSpace and ensured quality control for large-scale data acquisition in aerogeophysics.
 - **Scientific Output:** Co-authored book chapters, technical reports, and regional geological maps.

Education

- **2021 - 2025: PhD (Teknologie doktor) in Computer Science @ KTH Royal Institute of Technology**
 - **Thesis:** HPC on Cloud Systems, specializing in horizontal/vertical scaling and containerization.
 - Coordinated 3 Work Package deliverables and co-authored 1 for EU-funded Horizon projects OpenCUBE and IO-SEA.
- **2019 - 2021: Master's in Computer Science @ Universidade Federal da Bahia**
 - **Thesis:** Intelligent scheduling of search queries using Elasticsearch on DVFS processors.
 - **Relevant Courses:** Real-Time Systems, Artificial Intelligence, and Applied Statistics.
- **2010 - 2013: Bachelor's in Geophysics @ Universidade Federal do Rio Grande do Norte**
 - **Relevant Courses:** Geophysics of Petroleum Exploration, Seismic Data Processing, Geophysical Well logging, and Reservoir Geophysics. .
 - Elected Student Representative of the Geophysics Undergraduate Course Committee.

Leadership & Volunteering

- **Action10.org (2025):**
 - **Webmaster:** Fully redesigned the website using wordpress to enhance engagement with the potential audience and donors.
- **PhD Chapter @ KTH (Doktorandsektionen, 2022-2024):**
 - **University Board:** Represented 1,500+ doctoral students on the KTH University Board (*Universitetsstyrelsen*).
 - **Management:** Served as Chairperson, Auditor, and Webmaster; managed a **1M SEK** annual budget and a team of 10+ representatives.
 - **Digital Transformation:** Implemented a digital election platform, increasing student engagement by **7.5x**.

Technical Skills

- **Cloud & Orchestration:** Kubernetes, Kubeflow, Docker, Podman, Ceph Object Storage, Prometheus
- **HPC & Parallel Computing:** Slurm, MPI, OpenMP, CUDA/HIP, Parallel Computing Environments
- **Programming Languages:** Python, C/C++, Bash Scripting
- **Data & AI:** PyTorch, scikit-learn, OpenCV, MLflow, Pandas, NumPy, Matplotlib, Tableau
- **DevOps & Engineering:** Jenkins (CI/CD), Git, Terraform
- **Systems & Administration:** GNU/Linux, Debugging & Profiling (HPCToolkit, perf)
- **Geoscientific Suites:** Oasis montaj, SeisSpace, ArcGIS, GDAL, GeoPandas
- **Scientific Writing:** LaTeX, Markdown

Languages

- **English:** Fluent
- **Portuguese:** Native
- **Spanish:** Basic
- **Swedish:** Professional working knowledge

About me

I am a researcher with a unique technical trajectory spanning Geophysics and Computer Science. I am driven by how data-intensive systems and distributed algorithms can be engineered to solve high-stakes, real-world problems. During my PhD at KTH, I specialized in the intersection of HPC and cloud-native environments, focusing on making massive-scale systems more efficient and scalable.

At RISE, I translate these research insights into applied, industry-ready AI solutions, helping organizations navigate the complexities of modern computing infrastructure. I am equally comfortable debugging at the system level or architecting high-level AI workflows. I prioritize clarity, reproducibility, and pragmatic engineering—valuing solutions that are robust enough for production and flexible enough for research.

Publications

A complete and up-to-date list of my publications in Computer Science is available on dblp at <https://dblp.org/pid/298/8749.html>. The content related to when I worked as Geophysicist at the Geological Survey of Brazil can be found on <https://rigeo.sgb.gov.br>, with most of it being in portuguese.

This CV is designed to be ATS-friendly and as clear as possible with all the necessary relevant information for relevant parties.